Joseph DeRose

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EMPLOYMENT DETAILS

Tenure Track Assistant Staff Scientist 2024-present
Brookhaven National Laboratory

DESI Level 2 Manager for cross-analysis infrastructure; PI of BNL LDRD and DESI observing program on spectroscopy of weak lensing source galaxies.

Chamberlain Fellow 2021-2024
Lawrence Berkeley National Laboratory

Independent research program combining DESI spectroscopy with weak lensing and CMB lensing; co-chair of DESI cross-correlations working group; supervised UC Berkeley undergraduate and graduate students.

Postdoctoral Scholar 2019-2021
University of California, Santa Cruz & University of California, Berkeley

EDUCATION

PhD Physics 2014-2019
Stanford University

Thesis advisor: Risa Wechsler

BA Physics and Mathematics 2009-2013
University of California, Berkeley

FELLOWSHIPS AND AWARDS

- Chamberlain Prize Fellowship (5yr position), Lawrence Berkeley National Laboratory
- Paul Giddings Fellow, Kavli Institute for Particle Astrophysics and Cosmology
- Magna Cum Laude with honors in Physics and Mathematics, University of California, Berkeley

RESEARCH INTERESTS

- Joint analyses of spectroscopic and imaging surveys: DESI combined with weak lensing (DES, Rubin LSST) and CMB lensing
- Next-generation wide-field spectroscopy (DESI Run 2, Spec-S5): science cases, survey design, and spectroscopic calibration of weak lensing samples
- Large scale structure and CMB probes of dark energy, dark matter, neutrino mass, inflation and light relics
- Analytic (effective field theory) and simulation-based models of galaxy clustering, weak lensing, and the CMB
- High performance computing, machine learning, statistical inference

SCIENTIFIC COLLABORATIONS

- Dark Energy Survey: Builder 2014-present
- Rubin Observatory Dark Energy Science Collaboration: Member 2015-present
- Dark Energy Spectroscopic Instrument: Member 2015-present
- CMB S4: Member 2023-2025

LEADERSHIP POSITIONS

- Dark Energy Spectroscopic Instrument: Level 2 Manager – Cross Analysis Infrastructure Working Group 2024-present
- Rubin Observatory LSST Dark Energy Science Collaboration (DESC): DESC–DESI Run 2 Coordination Task Force Convener 2023-present
- Dark Energy Spectroscopic Instrument: Clusters, Cross-Correlations, and Small-scale Clustering Working Group Co-chair 2022-2024
- Dark Energy Survey Small Scales Analysis Team Convener 2017-2022

NEXT-GENERATION SPECTROSCOPY (DESI RUN 2 / SPEC-S5)

- PI, DESI Steel: dedicated DESI observations of weak lensing source galaxies to calibrate Rubin-like samples with wide-field spectroscopy 2026-2027
- PI, BNL LDRD: Mitigating systematics in Rubin LSST weak lensing source samples using wide-field spectroscopy 2027-2029
- Convener, DESC–DESI Run 2 Coordination Task Force: joint science case for DESI Run 2 and Rubin LSST 2023-present
- Contributor to Stage-5 spectroscopy planning (Snowmass 2021): *A spectroscopic road map for cosmic frontier: DESI, DESI-II, Stage-5* and *The MegaMapper* 2022

GRANTS AND COMPUTING ALLOCATIONS

- DOE/Brookhaven National Lab LDRD-B, "Mitigating Systematics in Rubin LSST Weak Lensing Source Galaxy Samples using Wide Field Spectroscopy", $250k/year, awarded (PI, 2027-2029).
- DESI, "DESI Steel", 12 hours DESI observing time ($100k equivalent) (PI, 2026-2027).
- INCITE, "AbacusFrontier: N-body simulations for a new generation of cosmic surveys", 510000 node hours (Co-I, 2026-2027).
- Fugaku, "Correlated Simulations for Joint Analysis of CMB and LSS", (Collaborator, 2023-2026).
- NASA ATP, "Leveraging Weak Gravitational Lensing - Redshift Space Distortions Cross-correlations", (Collaborator, 2022-2024).
- XSEDE, "N-body Simulations for Cosmic Acceleration and Neutrino Science with Wide Field Galaxy Surveys", 320000 node hours (Co-I, 3/22 - 3/23).
- XSEDE, "N-body Simulations for Cosmic Acceleration and Neutrino Science with Wide Field Galaxy Surveys", 43000 node hours (Co-I, 1/21 - 1/22).
- XSEDE, "Cosmological Simulations of Dark Energy and Massive Neutrinos for Wide Field Galaxy Surveys", 4000 node hours (PI, 6/20 - 6/21).

- NERSC, "Cosmological Simulations for Sky Surveys", 44k Node hours (co-I, 1/24 - 1/25).
- NERSC, "Cosmological Simulations for Sky Surveys", 25k Node hours (co-I, 1/23 - 1/24).
- NERSC, "Cosmological Simulations for Sky Surveys", 5M NERSC hours (co-I, 1/22 - 1/23).
- NERSC, "Cosmological Simulations for Sky Surveys", 5M NERSC hours (co-I, 1/21 - 1/22).
- NERSC, "Cosmological Simulations for Sky Surveys", 7M NERSC hours (co-I, 1/20 - 1/21).
- NERSC, "Cosmological Simulations for Sky Surveys", 7M NERSC hours (co-I, 1/19 - 1/20).
- NERSC, "Cosmological Simulations for Sky Surveys", 9M NERSC hours (co-I, 1/18 - 1/19).

SOFTWARE AND SIMULATION PRODUCTS

- **Aemulus** – Suite of N-body simulations and Gaussian-process/neural-network emulators for the matter and biased-tracer power spectra, halo mass function, halo bias, and galaxy clustering, including $w\nu$CDM cosmologies. [repo]
- **ADDGALS** – Algorithm for populating N-body simulations with realistic galaxy photometry, shapes, and SEDs to produce synthetic sky catalogs for wide-field surveys. [repo]
- **Buzzard / Cardinal** – Synthetic sky catalogs built on ADDGALS; widely used for validation of weak lensing, clustering, and 3×2pt analyses in DES, DESI and LSST DESC. Buzzard Simulations Cardinal Simulation
- **Zel'dovich Control Variates (ZCV)** – Variance-suppression technique for cosmological N-body simulations, yielding precision power-spectrum and clustering predictions at a fraction of the simulation cost.
- **HEFT Emulator** – Hybrid Effective Field Theory emulators based off the Aemulus simulations, applied to DES, DESI, BOSS, ACT and Planck. [repo]
- **gholax** – Machine learning accelerated, differentiable inference framework that uses neural network emulators to accelerate slow theory calculations and make them automatically differentiable. [repo]

SELECTED TALKS

30. DESC@KIPAC, KIPAC, Stanford, 12/25 (**Invited**)
29. APS Global Physics Summit 2025 – Latest Results from the Dark Energy Spectroscopic Instrument Special Session, , 3/25 (**Invited**)
28. Bay Area Strings, Information & Cosmology Symposium, , 11/24 (**Invited**)
27. Boston University Physics Seminar, Boston University, 3/24 (**Invited**)
26. Brookhaven National Laboratory Physics Seminar, BNL, 1/24 (**Invited**)
25. Berkeley Astrophysics Roundtable Donor Event, UC Berkeley, 12/23 (**Invited**)
24. SLAC Summer Institute: Artificial Intelligence in Fundamental Physics, SLAC, 8/23 (**Invited**)
23. Future Science with CMB x LSS , YITP, Kyoto, Japan, 4/23 (**Invited**)
22. Cosmology Colloquium, SLAC, 2/23 (**Invited**)
21. Cosmology Seminar, Kavli IPMU, 11/22 (**Invited**)

20. Intriguing Inconsistencies in the Growth of Structure over Cosmic Time, Sesto, Italy, 7/22 (**Invited**)
19. Berkeley Center for Cosmological Physics, Vipolze Conference, Smartno, Slovenia, 7/22
18. DES Y3 Highlight Plenary, DESI Collaboration Meeting, 6/22 (**Invited**)
17. Astro Seminar, NYU, 3/22 (**Invited**)
16. Astronomy Colloquium, University of British Columbia, 10/21 **(Invited)**
15. Astrophysics Colloquium, Stanford University, 6/21 **(Invited)**
14. Dark Energy Survey Year 3 Results Webinar, Virtual, 5/21
13. Institute for Nuclear and Particle Astrophysics Seminar, LBNL, 1/21 **(Invited)**
12. German Center for Cosmological Lensing Seminar, Remote talk, 5/20 **(Invited)**
11. Spectroscopic Surveys: Are We Ready For the Future?, UC Berkeley, 1/20 **(Invited)**
10. AAS Thesis Spotlight, Seattle, 1/19
9. FLASH Seminar, UC Santa Cruz, 9/18 **(Invited)**
8. Astro Seminar, NYU, 9/18 **(Invited)**
7. Berkeley Cosmology Seminar, UC Berkeley, 9/18 **(Invited)**
6. Modeling the Extragalactic Sky, UC Berkeley, 1/18 **(Invited)**
5. Astrophysics Colloquium, Stanford University, 9/17 **(Invited)**
4. Webinar, Laboratório Interinstitucional de e-Astronomia, Brazil, 7/17 **(Invited)**
3. COSMO16, University of Michigan, 8/16
2. KIPAC Tea, Stanford University, 9/16
1. Mock Santiago: Preparing for the Next Generation of Surveys, Universidad Catolica, Santiago, Chile, 4/16 **(Invited)**

STUDENTS/POSTDOCS (CO)SUPERVISED

- Vincent Su, B.S. Stanford ‘17
- Denise Lepore, B.S. CSU Pomona ‘19
- Amara McCune, B.S. Stanford ‘18 (now PhD student in physics at UC Santa Barbara)
- Duncan Wood, B.S. Stanford ‘17
- Judah Luberto, B.S. UC Santa Cruz ‘22
- Nishant Mishra, B.S. UC Berkeley ‘21 (now PhD student in astrophysics at University of Michigan)
- Shi-Fan Chen, Ph.D UC Berkeley ‘22 (now Hubble Fellow at Columbia University)
- Nickolas Kokron, Ph.D Stanford ‘23 (now postdoc at Princeton)
- Enia Xhakaj, Ph.D UC Santa Cruz ‘23
- Rose Hinson, B.S. UC Berkeley ‘24
- Danial Baradaran, B.S. UC Berkeley ‘25

- Jyotsna Ravi, B.S. UC Berkeley '26 (starting PhD at University of Michigan)
- Alexa Bartlett, Ph.D. UC Berkeley '28 (Expected)
- Hugo Camacho, Postdoc, BNL

SERVICE AND OUTREACH

- NSF AAG Panelist (2023)
- Astro Scholar Mentor, Berkeley Astronomy Department, 2020-2021
- LBNL INPA Seminar Committee 2021-2022
- DESI High coordinator 2021-2024
- DESI Education and Public Outreach Committee, 2021-2022
- Organizer for Astronomy on Tap (2016-2020)
- Lead Organizer for "Meetings of Astrophysics Students at Stanford" seminar series (2016-2018)
- Lead Organizer for Stanford Physics Computing Bootcamp (2016-2018)
- Referee: Astrophysical Journal, Monthly Notices of the Royal Astronomical Society, Journal of Cosmology and Astroparticle Physics, Astronomy & Astrophysics, Physical Review D

TEACHING

- Spring 2015: **PHYS25 Modern Physics**
  T.A.
- Winter 2017: **PHYS16 The Origin and Development of the Cosmos**
  T.A.
- Winter 2018: **PHYS16 The Origin and Development of the Cosmos**
  T.A.

REFERENCES

Risa Wechsler
Professor of Physics, Stanford University
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Martin White
Professor of Physics & Astronomy, UC Berkeley
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Alexie Leauthaud
Associate Professor of Astronomy, UC Santa Cruz
alexie@ucsc.edu

Bhuvnesh Jain
Professor of Physics, University of Pennsylvania
bjain@physics.upenn.edu

PUBLICATIONS

Papers: 193 Total citations: 18,000+ h-index: 69

Selected Lead-Author Publications

1. **J. DeRose** et al., *Steeling Weak Lensing Source Galaxy Samples against Systematics using Wide Field Spectroscopy* (2026). *arXiv preprint arXiv:2603.10113*. [1 citation]
2. **J. DeRose** and S. Chen, *The Lensing Counter Narrative: An Effective Description of Small-Scale Clustering in Weak Lensing Power Spectra* (2025). *arXiv preprint arXiv:2510.18981*. [7 citations]
3. S. Chen, **J. DeRose** et al., *Analysis of DESI×DES-Y3 using the Lagrangian effective theory of LSS* (2024). *Physical Review D*, 110(10), p. 103518. [42 citations]
4. **J. DeRose** et al., *Aemulus ν: precise predictions for matter and biased tracer power spectra in the presence of neutrinos* (2023). *Journal of Cosmology and Astroparticle Physics*, 2023(07), p. 054. [44 citations]
5. **J. DeRose** et al., *Precision redshift-space galaxy power spectra using Zel'dovich control variates* (2023). *Journal of Cosmology and Astroparticle Physics*, 2023(02), p. 008. [31 citations]
6. S. Pandey, E. Krause, **J. DeRose** et al., *Dark Energy Survey year 3 results: Constraints on cosmological parameters and galaxy-bias models from galaxy clustering and galaxy-galaxy lensing using the redMaGiC sample* (2022). *Physical Review D*, 106(4), p. 043520. [120 citations]
7. R. H. Wechsler, **J. DeRose** et al., *ADDGALS: Simulated sky catalogs for wide field galaxy surveys* (2022). *The Astrophysical Journal*, 931(2), p. 145. [56 citations]
8. **J. DeRose** et al., *Dark Energy Survey Year 3 results: Cosmology from combined galaxy clustering and lensing validation on cosmological simulations* (2022). *Physical Review D*, 105(12), p. 123520. [55 citations]
9. **J. DeRose** et al., *Neural network acceleration of large-scale structure theory calculations* (2022). *Journal of Cosmology and Astroparticle Physics*, 2022(04), p. 056. [30 citations]
10. **J. DeRose** et al., *Modeling redshift-space clustering with abundance matching* (2022). *The Astrophysical Journal*, 940(1), p. 13. [21 citations]
11. N. Kokron, **J. DeRose** et al., *The cosmology dependence of galaxy clustering and lensing from a hybrid N-body–perturbation theory model* (2021). *Monthly Notices of the Royal Astronomical Society*, 505(1), pp. 1422-1440. [103 citations]
12. **J. DeRose** et al., *The Aemulus project. I. Numerical simulations for precision cosmology* (2019). *The Astrophysical Journal*, 875(1), p. 69. [186 citations]
13. **J. DeRose** et al., *The buzzard flock: Dark energy survey synthetic sky catalogs* (2019). *arXiv preprint arXiv:1901.02401*. [135 citations]
14. N. MacCrann, **J. DeRose** et al., *DES Y1 Results: validating cosmological parameter estimation using simulated Dark Energy Surveys* (2018). *Monthly Notices of the Royal Astronomical Society*, 480(4), pp. 4614-4635. [49 citations]

Major Contributions / Mentored First Author

1. S. Chen, **J. DeRose** et al., *Cosmic Shear in Effective Field Theory at Two-Loop Order: Revisiting S_8 in Dark Energy Survey Data* (2026). *arXiv preprint arXiv:2603.28761*

2. N. Sailer, **J. DeRose** et al., *Evolution of structure growth during dark energy domination: Insights from the cross-correlation of DESI galaxies with CMB lensing and galaxy magnification* (2025). *Physical Review D*, 111(10), p. 103540. [13 citations]

3. A. E. Bayer, Y. Zhong, Z. Li, **J. DeRose** et al., *The HalfDome multi-survey cosmological simulations: N-body simulations* (2025). *Journal of Cosmology and Astroparticle Physics*, 2025(05), p. 016. [2 citations]

4. D. Shen, N. Kokron, **J. DeRose** et al., *Aemulus ν: precision halo mass functions in $w\nu$CDM cosmologies* (2025). *Journal of Cosmology and Astroparticle Physics*, 2025(03), p. 056. [2 citations]

5. A. Bartlett, **J. DeRose** et al., *Simulation budgeting for hybrid effective field theories* (2025). *arXiv preprint arXiv:2510.13962*.

6. N. Emas, A. Porredon, C. Blake, **J. DeRose** et al., *Validation of the DESI-DR1 3x2-pt analysis: scale cut and shear ratio tests* (2025). *arXiv preprint arXiv:2510.05539*.

7. S. Heydenreich, A. Leauthaud, **J. DeRose**. *Can dynamic dark energy explain the S_8 tension, the 'lensing is low' effect, or strong baryon feedback?* (2025). *arXiv preprint arXiv:2508.05746*.

8. E. Fernández-García, F. Prada, A. Smith, **J. DeRose** et al., *DESI DR2 reference mocks: clustering results from Uchuu-BGS and LRG* (2025). *arXiv preprint arXiv:2507.01593*.

9. N. Sailer et al., *Cosmological constraints from the cross-correlation of DESI Luminous Red Galaxies with CMB lensing from Planck PR4 and ACT DR6* (2025). *Journal of Cosmology and Astroparticle Physics*, 2025(06), p. 008. [36 citations]

10. K. Storey-Fisher, J. L. Tinker, Z. Zhai, **J. DeRose** et al., *The Aemulus Project. VI. Emulation of Beyond-standard Galaxy Clustering Statistics to Improve Cosmological Constraints* (2024). *The Astrophysical Journal*, 961(2), p. 208. [22 citations]

11. C. To, **J. DeRose** et al., *Buzzard to Cardinal: Improved Mock Catalogs for Large Galaxy Surveys* (2024). *The Astrophysical Journal*, 961(1), p. 59. [21 citations]

12. Z. Zhai, J. L. Tinker, A. Banerjee, **J. DeRose** et al., *The Aemulus Project. V. Cosmological Constraint from Small-scale Clustering of BOSS Galaxies* (2023). *The Astrophysical Journal*, 948(2), p. 99. [66 citations]

13. R. Zhou, S. Ferraro, M. White, **J. DeRose** et al., *DESI luminous red galaxy samples for cross-correlations* (2023). *Journal of Cosmology and Astroparticle Physics*, 2023(11), p. 097. [43 citations]

14. R. Ruggeri, C. Blake, **J. DeRose** et al., *A data compression and optimal galaxy weights scheme for Dark Energy Spectroscopic Instrument and weak lensing data sets* (2023). *Monthly Notices of the Royal Astronomical Society*, 525(3), pp. 3865-3878. [3 citations]

15. B. Hadzhiyska et al., *Mitigating the noise of DESI mocks using analytic control variates* (2023). *arXiv preprint arXiv:2308.12343*.

16. A. Amon et al., *Consistent lensing and clustering in a low-S8 Universe with BOSS, DES Year 3, HSC Year 1, and KiDS-1000* (2023). *Monthly Notices of the Royal Astronomical Society*, 518(1), pp. 477-503. [117 citations]

17. M. White, R. Zhou, **J. DeRose** et al., *Cosmological constraints from the tomographic cross-correlation of DESI Luminous Red Galaxies and Planck CMB lensing* (2022). *Journal of Cosmology and Astroparticle Physics*, 2022(02), p. 007. [149 citations]

18. S. Chen, M. White, **J. DeRose** et al., *Cosmological analysis of three-dimensional BOSS galaxy clustering and Planck CMB lensing cross correlations via Lagrangian perturbation theory* (2022). *Journal of Cosmology and Astroparticle Physics*, 2022(07), p. 041. [85 citations]

19. N. Kokron, S. Chen, M. White, **J. DeRose** et al., *Accurate predictions from small boxes: variance suppression via the Zel'dovich approximation* (2022). *Journal of Cosmology and Astroparticle Physics*, 2022(09), p. 059. [45 citations]

20. N. Kokron, **J. DeRose** et al., *Priors on red galaxy stochasticity from hybrid effective field theory* (2022). *Monthly Notices of the Royal Astronomical Society*, 514(2), pp. 2198-2213. [41 citations]

21. T. M. Abbott et al., *Dark Energy Survey Year 3 results: Cosmological constraints from galaxy clustering and weak lensing* (2022). *Physical Review D*, 105(2), p. 023520. [1380 citations]

22. A. Amon et al., *Dark Energy Survey Year 3 results: Cosmology from cosmic shear and robustness to data calibration* (2022). *Physical Review D*, 105(2), p. 023514. [535 citations]

23. J. Myles et al., *Dark Energy Survey Year 3 results: redshift calibration of the weak lensing source galaxies* (2021). *Monthly Notices of the Royal Astronomical Society*, 505(3), pp. 4249-4277. [232 citations]

24. J. L. Tinker, J. Cao, M. Alpaslan, **J. DeRose** et al., *Probing the galaxy–halo connection with total satellite luminosity* (2021). *Monthly Notices of the Royal Astronomical Society*, 505(4), pp. 5370-5388. [26 citations]

25. A. P. Hearin, N. Ramachandra, M. R. Becker, **J. DeRose**. *Differentiable predictions for large scale structure with shamnet* (2021). *arXiv preprint arXiv:2112.08423*. [19 citations]

26. S. Pandey et al., *Perturbation theory for modeling galaxy bias: Validation with simulations of the Dark Energy Survey* (2020). *Physical Review D*, 102(12), p. 123522. [61 citations]

27. M. Costanzi, C. To, **J. DeRose** et al., *Spectroscopic Quantification of Projection Effects in the SDSS redMaPPer Galaxy Cluster Catalog* (2020).

28. T. McClintock, E. Rozo, M. R. Becker, **J. DeRose** et al., *The Aemulus Project. II. Emulating the halo mass function* (2019). *The Astrophysical Journal*, 872(1), p. 53. [195 citations]

29. Z. Zhai, J. L. Tinker, M. R. Becker, **J. DeRose** et al., *The Aemulus project. III. Emulation of the galaxy correlation function* (2019). *The Astrophysical Journal*, 874(1), p. 95. [178 citations]

30. R. Buchs, C. Davis, D. Gruen, **J. DeRose** et al., *Phenotypic redshifts with self-organizing maps: A novel method to characterize redshift distributions of source galaxies for weak lensing* (2019). *Monthly Notices of the Royal Astronomical Society*, 489(1), pp. 820-841. [124 citations]

31. T. M. Abbott et al., *Dark Energy Survey year 1 results: Cosmological constraints from galaxy clustering and weak lensing* (2018). *Physical Review D*, 98(4), p. 043526. [1445 citations]

32. M. Gatti et al., *Dark Energy Survey Year 1 results: cross-correlation redshifts–methods and systematics characterization* (2018). *Monthly Notices of the Royal Astronomical Society*, 477(2), pp. 1664-1682. [105 citations]

33. D. Gruen, O. Friedrich, E. Krause, **J. DeRose** et al., *Density split statistics: Cosmological constraints from counts and lensing in cells in DES Y1 and SDSS data* (2018). *Physical Review D*, 98(2), p. 023507. [157 citations]

34. O. Friedrich, D. Gruen, **J. DeRose** et al., *Density split statistics: Joint model of counts and lensing in cells* (2018). *Physical Review D*, 98(2), p. 023508. [138 citations]

Collaborative Publications

1. N. Jeffrey et al., *Dark energy survey year 3 results: likelihood-free, simulation-based wCDM inference with neural compression of weak-lensing map statistics* (2025). *Monthly Notices of the Royal Astronomical Society*, 536(2), pp. 1303-1322.

2. L. Faga et al., *Dark energy survey year 3 results: cosmology from galaxy clustering and galaxy–galaxy lensing in harmonic space* (2025). *Monthly Notices of the Royal Astronomical Society*, 536(2), pp. 1586-1609.

3. S. Pandey et al., *Constraints on cosmology and baryonic feedback with joint analysis of Dark Energy Survey Year 3 lensing data and ACT DR6 thermal Sunyaev-Zel'dovich effect observations* (2025). *arXiv preprint arXiv:2506.07432*.

4. M. Abdul-Karim et al., *Data Release 1 of the Dark Energy Spectroscopic Instrument* (2025). *arXiv preprint arXiv:2503.14745*.

5. S. Heydenreich et al., *Lensing Without Borders: Measurements of galaxy-galaxy lensing and projected galaxy clustering in DESI DR1* (2025). *arXiv preprint arXiv:2506.21677*.

6. S. Bocquet et al., *Multiprobe cosmology from the abundance of SPT clusters and DES galaxy clustering and weak lensing* (2025). *Physical Review D*, 111(6), p. 063533.

7. J. Siegel et al., *Intrinsic alignment demographics for next-generation lensing: Revealing galaxy property trends with DESI Y1 direct measurements* (2025). *arXiv preprint arXiv:2507.11530*.

8. B. Thakore et al., *High-significance detection of correlation between the unresolved gamma-ray background and the large-scale cosmic structure* (2025). *Journal of Cosmology and Astroparticle Physics*, 2025(06), p. 037.

9. S. Sartori et al., *The imprint of cosmic voids from the DESI Legacy Survey DR9 Luminous Red Galaxies in the Planck 2018 lensing map through spectroscopically calibrated mocks* (2025).

10. A. Thomsen et al., *Dark Energy Survey Year 3 results: Simulation-based CDM inference from weak lensing and galaxy clustering maps with deep learning. I. Analysis design* (2025). *arXiv preprint arXiv:2511.04681*.

11. J. U. Lange et al., *A Unified Photometric Redshift Calibration for Weak Lensing Surveys using the Dark Energy Spectroscopic Instrument* (2025). *arXiv preprint arXiv:2510.25419*.

12. Q. Hang et al., *Biasing from galaxy trough and peak profiles with the DES Y3 redMaGiC galaxies and the weak lensing mass map* (2025). *arXiv preprint arXiv:2509.18967*.

13. R. Gomes et al., *Dark Energy Survey Year 3 Results: Cosmological constraints from second and third-order shear statistics* (2025). *arXiv preprint arXiv:2508.14018*.

14. S. Rauhut et al., *Testing gravitational physics by combining DESI DR1 and weak lensing datasets using the E_G estimator* (2025). *arXiv preprint arXiv:2507.16098*.

15. G. Zacharegkas et al., *Constraining the Stellar-to-Halo Mass Relation with Galaxy Clustering and Weak Lensing from DES Year 3 Data* (2025). *arXiv preprint arXiv:2506.22367*.

16. J. Prat et al., *Dark Energy Survey Year 3 results: CDM cosmology from simulation-based inference with persistent homology on the sphere* (2025). *arXiv preprint arXiv:2506.13439*.

17. M. Maus et al., *A joint analysis of 3D clustering and galaxy× CMB-lensing cross-correlations with DESI DR1 galaxies* (2025). *arXiv preprint arXiv:2505.20656*.

18. A. Adame et al., *The early data release of the dark energy spectroscopic instrument* (2024). *The Astronomical Journal*, 168(2), p. 58.

19. A. Adame et al., *Validation of the scientific program for the dark energy spectroscopic instrument* (2024). *The Astronomical Journal*, 167(2), p. 62.

20. S. Bocquet et al., *SPT clusters with DES and HST weak lensing. II. Cosmological constraints from the abundance of massive halos* (2024). *Physical Review D*, 110(8), p. 083510.

21. L. Bigwood et al., *Weak lensing combined with the kinetic Sunyaev–Zel'dovich effect: a study of baryonic feedback* (2024). *Monthly Notices of the Royal Astronomical Society*, 534(1), pp. 655-682.

22. M. Gatti et al., *Dark Energy Survey Year 3 results: Simulation-based cosmological inference with wavelet harmonics, scattering transforms, and moments of weak lensing mass maps. Validation on simulations* (2024). *Physical Review D*, 109(6), p. 063534.

23. S. Grandis et al., *The SRG/eROSITA All-Sky Survey-Dark Energy Survey year 3 weak gravitational lensing by eRASS1 selected galaxy clusters* (2024). *Astronomy & Astrophysics*, 687, A178.

24. S. Bocquet et al., *SPT clusters with DES and HST weak lensing. I. Cluster lensing and Bayesian population modeling of multiwavelength cluster datasets* (2024). *Physical Review D*, 110(8), p. 083509.

25. J. Kim et al., *The Atacama Cosmology Telescope DR6 and DESI: structure formation over cosmic time with a measurement of the cross-correlation of CMB lensing and luminous red galaxies* (2024). *Journal of Cosmology and Astroparticle Physics*, 2024(12), p. 022.

26. G. Giannini et al., *Dark Energy Survey Year 3 results: redshift calibration of the MagLim lens sample from the combination of SOMPZ and clustering and its impact on cosmology* (2024). *Monthly Notices of the Royal Astronomical Society*, 527(2), pp. 2010-2036.

27. M. Gatti et al., *Detection of the significant impact of source clustering on higher order statistics with DES Year 3 weak gravitational lensing data* (2024). *Monthly Notices of the Royal Astronomical Society: Letters*, 527(1), pp. L115-L121.

28. J. McCullough et al., *DESI complete calibration of the colour–redshift relation (DC3R2): results from early DESI data* (2024). *Monthly Notices of the Royal Astronomical Society*, 531(2), pp. 2582-2602.

29. J. U. Lange et al., *Systematic Effects in Galaxy-Galaxy Lensing with DESI* (2024). *arXiv preprint arXiv:2404.09397*.

30. S. Yuan et al., *Redshift evolution and covariances for joint lensing and clustering studies with DESI Y1* (2024). *Monthly Notices of the Royal Astronomical Society*, 533(1), pp. 589-607.

31. J. McCullough et al., *Dark energy survey year 3: Blue shear* (2024). *arXiv preprint arXiv:2410.22272*.

32. C. Blake et al., *The DESI-Lensing Mock Challenge: large-scale cosmological analysis of 3x2-pt statistics* (2024). *arXiv preprint arXiv:2412.12548*.

33. N. Chicoine et al., *Weak Gravitational Lensing around Low Surface Brightness Galaxies in the DES Year 3 Data* (2024). *arXiv preprint arXiv:2407.19081*.

34. A. Campos et al., *Enhancing weak lensing redshift distribution characterization by optimizing the Dark Energy Survey Self-Organizing Map Photo-z method* (2024). *arXiv preprint arXiv:2408.00922*.

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